Task 14 — Fairness, Bias Audit & Explainability

Data Analyst · Sprint C - Intelligence Layer · Task Brief

DATA ANALYST PlaceMux · Phase 3 · Task Brief

TASK 14 / 25 **Enterprise Grade · Expert** Sprint C - Intelligence Layer · Mastery 82%

Focus

Run the fairness and bias analysis: measure outcome disparities across groups and defend the methodology.

What to build / complete

- An outcome-disparity analysis across relevant groups
- A defensible methodology (metric choice, proxies, limitations, consent)
- Monitoring so fairness is tracked continuously, not audited once

Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: You can state, with evidence and honest caveats, whether PlaceMux treats groups comparably.

Stage B — Build: An outcome-disparity analysis across relevant groups

1. Define the metric, its source & the decision

Define precisely what “An outcome-disparity analysis across relevant groups” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage C — Build: A defensible methodology (metric choice, proxies, limitations, consent)

1. Define the metric, its source & the decision

Define precisely what “A defensible methodology (metric choice, proxies, limitations, consent)” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage D — Build: Monitoring so fairness is tracked continuously, not audited once

1. Define the metric, its source & the decision

Define precisely what “Monitoring so fairness is tracked continuously, not audited once” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Present the disparity analysis with its methodology, limitations and ongoing monitoring.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

Done when (demoable by end of task)

- “An outcome-disparity analysis across relevant groups” is complete, real, and demoable end-to-end.
- “A defensible methodology (metric choice, proxies, limitations, consent)” is complete, real, and demoable end-to-end.
- “Monitoring so fairness is tracked continuously, not audited once” is complete, real, and demoable end-to-end.
- Good looks like: You can state, with evidence and honest caveats, whether PlaceMux treats groups comparably.

Worry if any of these are true

- Fairness claimed without measurement.
- A one-off audit with no ongoing monitoring.
- Calling an experiment early because it looked good on day two.

Hand-off

- Fairness findings to AI-ML and Compliance. Make sure the next person can pick it up without you.